

PHETCHABUN, Thailand

WMO: 483790

Lat: 16.4347N

Lon: 101.1517E

Elev: 116

StdP: 99.94

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	14.9	16.5	9.2	7.4	19.9	11.3	8.5	21.6	5.4	24.7	4.8	25.1	0.1	10	0.308

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	11.5	38.5	26.2	37.3	26.0	36.2	25.8	28.1	34.3	27.6	33.5	27.2	32.8	1.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.6	22.5	31.3	26.2	21.9	30.5	25.9	21.5	30.0	91.2	34.4	89.0	33.8	87.4	33.1	31.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
(4) 4.4	3.8	3.3	12.1	40.1	2.2	1.0	10.5	40.8	9.2	41.4	7.9	42.0	6.3	42.7		
(5)			10.6	29.2	2.1	1.1	9.1	29.9	7.8	30.5	6.6	31.1	5.1	31.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.0	25.2	27.3	29.4	30.8	29.8	29.3	28.4	28.1	28.1	28.0	26.8	25.0
	DBStd	2.42	2.14	2.12	2.07	1.94	1.76	1.38	1.29	1.11	1.08	1.25	1.86	2.37
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	1	0	0	0	0	0	0	0	0	0	0	0	1
	CDD10.0	6569	472	483	603	623	614	578	569	560	543	557	503	464
	CDD18.3	3528	214	250	344	373	356	328	311	302	293	298	253	207
	CDH23.3	38449	2037	2709	4206	4793	4203	3827	3346	3087	2907	2971	2455	1909
CDH26.7	17525	904	1407	2344	2708	2023	1708	1297	1082	1003	1181	1071	797	
Wind	WSAvg	0.6	0.5	0.5	0.7	0.7	0.7	0.7	0.8	0.6	0.5	0.6	0.7	0.7
Precipitation	PrecAvg	1149	9	14	38	73	157	134	180	212	213	102	15	3
	PrecMax	1774	67	44	116	185	284	279	291	344	367	278	64	19
	PrecMin	796	0	0	0	0	60	62	99	56	106	17	0	0
	PrecStd	258	21	14	32	48	61	65	59	68	86	68	15	6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.1	37.2	39.2	40.7	38.9	37.0	35.6	34.2	34.2	34.8	35.2	34.8
		MCWB	23.3	24.5	25.1	26.8	27.2	26.6	26.5	27.1	26.8	25.8	25.5	24.2
	2%	DB	33.8	35.9	37.9	39.1	37.0	35.2	33.8	33.1	33.1	33.7	34.1	33.4
		MCWB	23.1	24.2	25.2	26.5	27.2	26.8	26.8	27.0	26.6	25.8	25.0	23.6
	5%	DB	32.8	34.9	36.8	37.7	35.5	34.0	32.7	32.1	32.2	32.9	33.1	32.3
		MCWB	22.8	24.1	25.0	26.5	27.1	26.8	26.6	26.7	26.5	25.4	24.5	23.1
	10%	DB	31.6	33.7	35.7	36.3	34.0	32.9	31.8	31.2	31.2	32.0	32.0	31.0
		MCWB	22.3	23.8	24.9	26.3	26.9	26.7	26.4	26.5	26.3	25.3	23.9	22.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.7	26.7	27.3	28.8	28.9	28.2	28.1	27.9	27.8	27.3	26.9	25.8
		MCDB	32.3	34.7	36.0	37.3	35.9	33.2	32.5	32.5	32.5	32.5	33.0	32.1
	2%	WB	24.7	25.9	26.6	27.7	28.1	27.6	27.4	27.3	27.3	26.8	26.1	24.9
		MCDB	31.2	33.9	35.4	36.4	34.5	32.8	31.9	31.7	31.6	31.6	31.9	31.0
	5%	WB	24.0	25.2	26.0	27.2	27.5	27.2	27.0	27.0	26.9	26.4	25.5	24.1
		MCDB	30.0	32.9	34.8	35.7	33.7	32.3	31.6	31.2	31.0	30.9	30.8	30.0
	10%	WB	23.2	24.5	25.5	26.6	27.1	26.9	26.6	26.6	26.5	25.9	24.8	23.3
		MCDB	29.1	31.7	33.9	34.7	33.0	31.8	30.8	30.4	30.3	30.2	29.7	29.2
Mean Daily Temperature Range	5% DB	MDBR	13.4	13.7	13.0	11.5	8.9	7.9	7.0	6.8	7.1	8.6	10.7	12.4
		MCDBR	14.7	15.0	14.7	13.1	11.0	9.3	8.4	8.0	8.3	9.9	11.7	13.3
		MCWBR	5.5	5.4	5.0	4.2	3.5	2.9	3.0	3.0	3.0	3.2	3.8	4.7
	5% WB	MCDBR	12.2	13.3	13.1	12.0	9.7	8.3	7.6	7.4	7.6	8.6	10.4	11.9
		MCWBR	5.0	5.1	4.6	4.2	3.6	3.1	3.1	3.1	3.0	3.1	3.7	4.6
Clear-Sky Solar Irradiance	taub	0.486	0.591	0.671	0.619	0.505	0.453	0.442	0.451	0.491	0.538	0.464	0.465	
	taud	2.073	1.829	1.683	1.837	2.182	2.349	2.377	2.336	2.209	2.036	2.228	2.160	
	Ebn at Noon	798	732	689	728	805	839	851	851	819	766	815	806	
	Edh at Noon	158	211	251	215	150	126	123	129	146	169	134	141	
All-Sky Solar Radiation	RadAvg	4.72	5.13	5.29	5.70	5.36	5.04	4.68	4.51	4.49	4.69	4.84	4.63	
	RadStd	0.32	0.24	0.33	0.28	0.32	0.35	0.27	0.24	0.27	0.33	0.22	0.17	

Historical Trends

		Heating		Cooling		Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	+0.45	N/A	N/A	N/A	+0.27	N/A	N/A	N/A	+209	+212	
(47) Regional (20 neighbors)	N/A	N/A	N/A	N/A	+0.14	+0.16	N/A	N/A	+106	+107	

Nomenclature: See separate page